Signals and Systems — Reference

how to read this note

Every method is a chain of steps. Each step is a template with empty colored slots.

Do	How
1. find the pieces	Each topic opens with a PIECES table. It says where each piece is in your problem, and its color.
2. fill the slots	A piece goes in the slot of the same color: ω goes in $\boxed{\omega}$.
3. one box at a time	Each box is one move. Do not merge two boxes.

Sinusoid colors stay fixed: amplitude, frequency, phase, sampling rate. Other topics set their colors in their PIECES tables. The number in parentheses in the tables below is the topic page.

the big picture

Everything in this course is one of two things:

- a **signal**: a quantity that changes over time, like a voltage or a sound.
- a **system**: a machine that takes a signal in and puts a signal out.

The topics build on each other in this order:

1. **How big is a signal?** Energy and power (4–5).
2. **Building blocks.** The button δ, the switch u, and the ramp r (10).
3. **Machines.** Systems, block diagrams, difference equations (11).
4. **Describing machines.** Memory, causality, stability, linearity, time invariance (12).
5. **One echo describes the machine.** The impulse response h and the convolution sum $y = x * h$ (13).

The link from 1 to 4: BIBO stability asks if a *bounded* input gives a *bounded* output. That is a question about signal size.

plain-English glossary

Symbol	Say it as
$\delta[n]$, $\delta(t)$	**the button**: pressed once at time 0, then released
$u[n]$, $u(t)$	**the switch**: flipped on at time 0 and left on
$r[n]$, $r(t)$	**the ramp**: the running total of the switch, $0, 1, 2, 3, \ldots$
t	the time variable. It sweeps through every time, and is gone after you integrate
T	the half-width of a window $[-T, T]$. It is the thing that goes to ∞
t_0, n_0	"t-naught": one fixed time, such as 3
N	DT window half-width ($2N+1$ samples), or a DT period
ε	"epsilon": a tiny number that we shrink toward 0
$\int \ldots dt$	add up area as t sweeps (CT)
$\sum$	add up separate terms (DT), the DT twin of $\int$
dt	an infinitely thin slice of time. Not Δt.
B	a ceiling: the input never goes above B in size
$h[n]$	**the echo** (impulse response): what an LTI machine puts out for one button press
$x * h$	"x convolved with h: add up one echo per input stem

notation: CT and DT side by side

Item	Continuous time (CT)	Discrete time (DT)
signal	$x(t)$, round brackets	$x[n]$, square brackets
time variable	t, any real number	n, integers only
period	T (seconds)	N (samples, an integer)
frequency	ω (rad/s), f (Hz)	$\hat{\omega}$ (rad/sample)
energy, power	integral $\int$	sum $\sum$
rate of change	derivative $\frac{d}{dt}$	first difference $x[n] - x[n-1]$
running total	integral $\int_{-\infty}^{t}$	running sum $\sum_{k=-\infty}^{n}$
button	$\delta(t)$, area 1	$\delta[n]$, height 1

sinusoid quantities

Symbol	Name	Unit	Link
A	amplitude	same as x	peak value
ω	angular frequency	rad/s	$\omega = 2\pi f$
f	frequency	Hz = cycles/s	$f = \frac{1}{T}$
T	period	s	$T = \frac{2\pi}{\omega}$
ϕ	phase	rad	$\phi = -\omega t_0$
t_0	time shift	s	$t_0 = -\frac{\phi}{\omega}$
$\hat{\omega}$	DT frequency	rad/sample	$\hat{\omega} = \omega T_s$
T_s, f_s	sampling period, rate	s, Hz	$f_s = \frac{1}{T_s}$

trig and complex number facts

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

$\sin 0 = 0$ and $\cos 0 = 1$. Do not swap them.

sine as a cosine	$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$
negative amplitude	$-A\cos\theta = A\cos(\theta + \pi)$
square of a cosine	$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$
Euler's formula	$e^{j\theta} = \cos\theta + j\sin\theta$
cosine from exponentials	$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$
sine from exponentials	$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$
size of $a + jb$	$\sqrt{a^2 + b^2}$
angle of $a + jb$	$\arctan\frac{b}{a}$ (check the quadrant)
polar form	$C = \lvert C\rvert e^{j\phi}$
size of $e^{j\theta}$	1
whole turns	$e^{j2\pi k} = 1$ for each integer k

sums, integrals, limits

Item	Result
infinite geometric sum, $\lvert r\rvert < 1$	$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$
finite geometric sum, $r \neq 1$	$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$
count of terms from $-N$ to N	$2N+1$
$\int t^n\, dt$	$\frac{t^{n+1}}{n+1}$
$\int e^{at}\, dt$	$\frac{e^{at}}{a}$
$\int_0^{\infty} e^{-at}\, dt, \quad a > 0$	$\frac{1}{a}$
definite integral	$F(\text{top}) - F(\text{bottom})$
fixed number over a growing T	$\lim_{T\to\infty} \frac{c}{T} = 0$
a growing T alone	$\lim_{T\to\infty} cT = \infty \quad (c > 0)$
T cancels	$\frac{cT}{2T} = \frac{c}{2}$, a constant

system property tests

Property	Ask yourself	Yes / no example
memoryless	Does y now use only x now?	yes x^2 no $x[n-1]$
causal	Does y avoid peeking at the future?	yes $x[n-1]$ no $x[n+1]$
stable	Does a capped input give a capped output?	yes x^2 no $\sum_{k\le n} x[k]$
invertible	Can I undo it and get x back?	yes $2x$ no x^2
time invariant	Does it ignore the clock? Delay in = delay out	yes $x[n-1]$ no $nx[n]$
linear	Scale in → scale out, add in → add out?	yes $nx[n]$ no $2x+3$

properties of common DT systems

System	Memoryless	Causal	Linear	Time inv.	Stable
$x[n-1]$, $x[n]-x[n-1]$, moving average	no	yes	yes	yes	yes
accumulator	no	yes	yes	yes	no
$nx[n]$	yes	yes	yes	no	no
$x[n]^2$, $2x[n]+3$	yes	yes	no	yes	yes
$x[n+1]$	no	no	yes	yes	yes
$x[2n]$, $x[-n]$	no	no	yes	no	yes
$0.5y[n-1]+x[n]$ (at rest)	no	yes	yes	yes	yes

which tool? — given → find

Given → find	Do this	
$x(t)$, $T_s \to x[n]$	put $t = nT_s$	(1)
$A\cos(\omega t+\phi) \to f, T, t_0$	read, then divide	(2)
$f, f_s \to \hat{\omega}$	$\hat{\omega} = 2\pi f/f_s$	(3)
$f, f_s \to$ aliased?	Nyquist: $f_s > 2f$	(3)
dies out or finite length	energy E_∞	(4)
periodic or constant	power P_∞	(4)
a graph → energy	square each piece, add	(5)
$x(t) \to x(at+b)$	map each breakpoint	(6)
$y = x(at+b) \to x$	$x(t) = y((t-b)/a)$	(6)
two graphs → $y = Ax(at+b)$	match two points, solve	(6)
signal → period	$2\pi/\omega$, or reduce $\hat{\omega}/2\pi$	(7)
signal → even, odd, neither	compare $x(-t)$ with $x(t)$	(8)
$Ce^{at} \to$ real signal	Euler, then the real part	(9)
integral with $\delta(t-t_0)$	the flash: three questions	(10)
a pulse graph → formula	a switch on, a switch off	(10)
DT stems → buttons	one $x[k]\delta[n-k]$ per stem	(10)
graph with corners → formula	a ramp per corner	(10)
blocks → one equation	substitute block by block	(11)
equation ↔ block diagram	one D per shift	(11)
system → properties	ask each question	(12)
system → stable?	cap the input with B	(12)
system → inverse	solve for x, inverse table	(12)
equation → h	read coefficients, or run $x = \delta$	(13)
$h \to$ causal, stable	$h[n<0] = 0$, $\sum\lvert h\rvert$	(13)
stems x, $h \to y$	table: one row per stem	(13)
pulse and formula → y	flip and slide, by regions	(13)
two formulas → y	sum $k = 0..n$, geometric	(13)

The number in parentheses is the topic page.

1 — signals and sampling

THE IDEA
A **signal** changes over time. **Sampling** takes one snapshot every T_s seconds: the movie $x(t)$ becomes a stack of photos $x[n]$.

EXPLANATION

Pieces

Piece	Where to find it
$x(t)$	the CT signal (the movie)
T_s	seconds between photos. Or $T_s = \frac{1}{f_s}$ from the rate
$x[n]$	the DT signal (the photos)

CT	DT
$x(t)$, round brackets	$x[n]$, square brackets
t any real number	n an integer only

Trap: $x[2.5]$ does not exist. Between samples a DT signal is undefined, not zero.
Trap: T_s is seconds per photo. f_s is photos per second. Flip one to get the other.

WORK

Steps — CT → DT (sample)

1. READ THE PIECES
$x(t)$ T_s *(if given f_s: $T_s = 1/f_s$)*

↓

2. SUBSTITUTE
$x[n] = \boxed{x}\left(n \cdot \boxed{T_s}\right)$ *(write nT_s wherever the formula says t)*

↓

3. SIMPLIFY
Collect the numbers. The answer has only n in it.

Steps — DT → CT (undo)

1. READ THE PIECES
$x[n]$ T_s

↓

2. SOLVE FOR n
photo n was taken at $t = nT_s$, so $n = \dfrac{t}{\boxed{T_s}}$

↓

3. SUBSTITUTE
$x(t) = \boxed{x}\left[\dfrac{t}{\boxed{T_s}}\right]$ *(write t/T_s wherever the formula says n)*

2 — sinusoids

THE IDEA
A dot on a spinning wheel. Its height over time is a sinusoid. Wheel size: amplitude. Spin rate: frequency. Where the dot starts: phase.

EXPLANATION

Pieces

Piece	Where to find it
A	the number in front of cos. The peak
ω	the number multiplying t (rad/s)
ϕ	the number added inside, sign included (rad)

Exact form first: $x(t) = A\cos(\omega t + \phi)$.

If you have	Rewrite as
$A\sin\theta$	$A\cos\left(\theta - \frac{\pi}{2}\right)$
$-A\cos\theta$	$A\cos(\theta + \pi)$

Which way?

Given → find	Do this
$\omega \to f$	$f = \dfrac{\omega}{2\pi}$
$f \to \omega$	$\omega = 2\pi f$
$f \to T$ / $T \to \omega$	$T = \dfrac{1}{f}$ / $\omega = \dfrac{2\pi}{T}$
$\phi \to t_0$	$t_0 = -\dfrac{\phi}{\omega}$
$t_0 \to \phi$	$\phi = -\omega t_0$

Quick example	Work → answer
$3\cos(4\pi t - \frac{\pi}{3})$	$A = 3$, $\omega = 4\pi$, $\phi = -\frac{\pi}{3}$
$4\sin(10t)$	$= 4\cos(10t - \frac{\pi}{2})$: 4, 10, $-\frac{\pi}{2}$

Trap: Negative ϕ means $t_0 > 0$: delayed, moves right.
Trap: The first peak is at $t = t_0$. That is the shift.
Trap: Keep ϕ in $(-\pi, \pi]$: add or subtract 2π.

WORK

Steps — read the parts

1. EXACT FORM
Rewrite to $\boxed{A}\cos\left(\boxed{\omega}t + \boxed{\phi}\right)$. Read the three pieces.

↓

2. FREQUENCY
$f = \dfrac{\boxed{\omega}}{2\pi}$ Hz *(one turn is 2π rad)*

↓

3. PERIOD
$T = \dfrac{1}{f} = \dfrac{2\pi}{\boxed{\omega}}$ s

↓

4. TIME SHIFT
$t_0 = -\dfrac{\boxed{\phi}}{\boxed{\omega}}$ s *($A\cos(\omega(t - t_0))$)*

Steps — build from the parts

1. AMPLITUDE
A = the peak height.

↓

2. FREQUENCY
$\boxed{\omega} = \dfrac{2\pi}{T}$ or $2\pi f$.

↓

3. PHASE
t_0 = time of the first peak. $\boxed{\phi} = -\boxed{\omega}t_0$

↓

4. ASSEMBLE
$x(t) = \boxed{A}\cos\left(\boxed{\omega}t + \boxed{\phi}\right)$

3 — DT sinusoids, DT frequency, aliasing

THE IDEA
Photograph the spinning wheel with a strobe. In the photos you only see how far it turned between photos. That angle per photo is the **DT frequency** $\hat{\omega}$ (rad/sample).

EXPLANATION

Pieces

Piece	Where to find it
ω or f	the CT frequency (rad/s or Hz)
f_s or T_s	the sampling rate or period
$\hat{\omega}$	the DT frequency: the number multiplying n

Given → find	Do this
$f, f_s \to \hat{\omega}$	$\hat{\omega} = \dfrac{2\pi f}{f_s}$
$\omega, T_s \to \hat{\omega}$	$\hat{\omega} = \omega T_s = \dfrac{\omega}{f_s}$
$\hat{\omega}, f_s \to f$	$f = \dfrac{\hat{\omega} f_s}{2\pi}$ (the lowest answer)

Aliasing: $\hat{\omega}$, $\hat{\omega} + 2\pi k$ and $-\hat{\omega}$ give the same photos. In Hz: $\pm f + k f_s$ all look the same. **Nyquist**: no aliasing if $f_s > 2 f_{max}$. The frequency you see after sampling is in $[0, \frac{f_s}{2}]$.
Trap: DT → CT has more than one answer. Any alias fits.
Trap: cos is even, so $f_s - f$ aliases to f too.

WORK

Steps — CT → DT frequency

1. READ THE PIECES
ω (or f) $\quad f_s$

↓

2. DIVIDE
$$\boxed{\hat{\omega}} = \frac{\boxed{\omega}}{\boxed{f_s}} = \frac{2\pi \boxed{f}}{\boxed{f_s}}$$

↓

3. WRITE $x[n]$
$x[n] = A\cos\left(\boxed{\hat{\omega}} n + \phi\right)$ *(A and ϕ do not change)*

Steps — DT → CT frequency

1. READ THE PIECES
$\hat{\omega} \quad f_s$

↓

2. MULTIPLY
$$f = \frac{\boxed{\hat{\omega}} \cdot \boxed{f_s}}{2\pi}$$ *(the lowest answer. Others: $\pm f + k f_s$)*

Steps — does f alias to f'?

1. DT FREQUENCY
$$\boxed{\hat{\omega}} = \frac{2\pi \boxed{f}}{\boxed{f_s}}$$

↓

2. REMOVE WHOLE TURNS
Subtract 2π until $\hat{\omega}$ is in $(-\pi, \pi]$.

↓

3. COMPARE
Same $|\hat{\omega}|$ as $f' \Rightarrow$ same photos (aliases).

Steps — the frequency you actually see

1. NYQUIST
$\boxed{f_s} > 2\boxed{f}$? Yes ⇒ you see f itself. Stop.

↓

2. REMOVE WHOLE f_s
Subtract $\boxed{f_s}$ from $\boxed{f}$ until it is in $(-\frac{f_s}{2}, \frac{f_s}{2}]$.

↓

3. DROP THE SIGN
The apparent frequency is $|$that$|$. *(cos is even)*

4 — energy and power

THE IDEA
Think of $|x|^2$ as money spent per day, forever. **Energy** is the total. **Power** is the average per day. Spend \$100 once: total finite, average 0. Spend \$5 a day: total ∞, average \$5.

EXPLANATION

Pieces

Piece	Where to find it
on at	where the signal starts. A switch $u(t-a)$ turns on at a
off at	where it ends (∞ if never)
$\lvert x\rvert^2$	square the signal first
T (or N)	the window half-width. It goes to ∞

Classify by shape first

The signal	Result
dies out, or finite length	**energy signal**: E finite, $P=0$
periodic, or constant	**power signal**: P finite, $E=\infty$
grows without limit ($x=t$)	**neither**: both ∞

Shortcuts

Signal	Result
sinusoid, amplitude A	$P=\frac{A^2}{2}$
constant c	$P=c^2$
$Ae^{-at}u(t)$, $a>0$	$E=\frac{A^2}{2a}$
$a^n u[n]$, $\lvert a\rvert<1$	$E=\frac{1}{1-a^2}$
switch $u(t)$ or $u[n]$	$P=\frac{1}{2}$ *(on for half the window)*
$Ce^{j\omega t}$	$P=\lvert C\rvert^2$
$c+\sum A_i\cos(\omega_i t+\phi_i)$, different ω_i	$P=c^2+\sum\frac{A_i^2}{2}$

Trap: Square first, then add. Negatives count positive.
Trap: "16 forever" is the power. The energy is ∞.
Trap: A final E or P never contains T or N. $\frac{c}{T}\to 0$.
Trap: Write the label: energy signal, power signal, or neither.

WORK

Steps — energy

1. CLASSIFY
Use the shape table. Periodic or constant → go to power.

↓

2. READ THE PIECES
on at, off at, and $\lvert x\rvert^2$

↓

3. INTEGRATE (DT: SUM)
$$E=\int_{\text{on}}^{\text{off}}\lvert x\rvert^2\,dt \qquad E=\sum_{\text{on}}^{\text{off}}\lvert x\rvert^2$$
(where $x=0$ it adds nothing, so the limits shrink)

↓

4. LABEL
E finite $\Rightarrow P=\lim\frac{E}{2T}=0\Rightarrow$ energy signal

Steps — power

1. WINDOW TOTAL
$$\int_{-T}^{T}\lvert x\rvert^2\,dt \qquad \text{DT: } \sum_{-N}^{N}\lvert x\rvert^2$$

↓

2. DIVIDE BY THE WINDOW
CT: $\div\,2T$ DT: $\div\,(2N+1)$

↓

3. LIMIT
Let $T\to\infty$. T must cancel or vanish.

↓

4. LABEL
P finite, nonzero $\Rightarrow E=\infty\Rightarrow$ power signal

Steps — periodic shortcut

1. ONE PERIOD
$$P=\frac{1}{T_0}\int_{\text{one period}}\lvert x\rvert^2\,dt$$ *(DT: $\frac{1}{N}\sum$ over one period)*

5 — energy from a graph

THE IDEA
Do not write one big integral. Chop the graph into straight pieces, find each piece's energy from the shortcut table, and add. In DT there are no widths: each stem's energy is its height squared.

EXPLANATION

Pieces (one per straight piece)

Piece	Where to find it
A	the height at the start of the piece
B	the height at the end of the piece
D	the length (duration) of the piece
h	DT: the height of one stem

Shortcut table (the templates)

Piece of the graph	Energy of that piece
flat at level A	A^2D
ramp $0 \to A$ (or $A \to 0$)	$\dfrac{A^2D}{3}$
line $A \to B$	$\dfrac{D}{3}\left(A^2+AB+B^2\right)$
DT stem of height h	h^2

Line through (t_1, x_1) with slope m: $x = x_1 + m(t - t_1)$, if you want to check by integral.
Trap: A squared ramp is a curve: $\div 3$, not $\div 2$.
Trap: Negative heights count positive: $(-2)^2 = 4$.
Trap: If the graph repeats forever, $E = \infty$. Find the power over one period.

WORK

Steps — energy from a graph

1. CHOP
Split the graph at every corner and jump. Each piece is flat, a ramp, or a line.

↓

2. READ EACH PIECE
A, B, D for each piece. DT: each h.

↓

3. SHORTCUT EACH PIECE
flat: $\boxed{A}^2\boxed{D}$ ramp: $\dfrac{\boxed{A}^2\boxed{D}}{3}$ line: $\dfrac{\boxed{D}}{3}\left(\boxed{A}^2+\boxed{A}\boxed{B}+\boxed{B}^2\right)$

↓

4. ADD
$E =$ sum of the pieces. DT: $E = \sum \boxed{h}^2$

↓

5. LABEL
Finite length ⇒ energy signal.

Steps — a repeating graph (power)

1. ONE PERIOD
Find the period T_0. Chop one period only.

↓

2. ENERGY OF ONE PERIOD
Steps 2–4 above, over one period.

↓

3. DIVIDE
$P = \dfrac{E_{\text{one period}}}{T_0}$ *(DT: divide by N, the samples in one period)* ⇒ power signal

6 — transformations of the time variable

THE IDEA
The shape does not change, only *when* things happen. Ask each corner of the graph one question: **where did this point move to?** It keeps its height and lands at a new time.

Sketch $x(at+b)$ from $x(t)$

EXPLANATION

Pieces

Piece	Where to find it
t_{old}	each corner or jump of the $x(t)$ graph
a	the number multiplying t inside
b	the number added inside, sign included

Inside	Effect
$x(t-t_0)$, $t_0>0$	moves right (delay)
$x(t+t_0)$, $t_0>0$	moves left (advance)
$x(-t)$	mirror about $t=0$
$x(at)$, $\lvert a\rvert>1$ / <1	compress / stretch

Trap: The sign inside is opposite to the direction: $x(t-3)$ moves right.
Trap: In two moves: shift by b first ($x(t+b)$), then scale by a. Scale-then-shift by b is wrong. The point map avoids the order.
Trap: DT: $x[2n]$ keeps only the even samples. It cannot be undone.

WORK

1. LIST THE CORNERS
Each corner t_{old} of $x(t)$, with its height.

↓

2. READ a, b
Match the inside to $\boxed{a}t+\boxed{b}$.

↓

3. MAP EACH CORNER
$$t_{\text{new}}=\frac{\boxed{t_{\text{old}}}-\boxed{b}}{\boxed{a}}$$ *(heights do not change)*

↓

4. DRAW
Connect the moved corners in the same order. If $a<0$ the order reverses.

↓

5. DT ONLY
Keep only integer t_{new}. A non-integer sample is lost.

Undo, or find $y=Ax(at+b)$ from two graphs

EXPLANATION

Pieces

Piece	Where to find it
A	height of y ÷ height of x. Upside down: $A<0$
a	negative if y is a mirror of x
(t_1,τ_1), (t_2,τ_2)	two matching points: x at t_1 moved to τ_1 in y

Given → find	Do this
$x(t)\to x(at+b)$	map each corner (above)
$y(t)=x(at+b)\to x$	$x(t)=y\left(\frac{t-b}{a}\right)$
two graphs $\to A,a,b$	two matching points (right)

WORK

Steps — undo $y=x(at+b)$

1. TEMPLATE
$$x(t)=y\left(\frac{t-\boxed{b}}{\boxed{a}}\right)$$ *(solve $at+b=s$ for t, rename $s\to t$)*

Steps — find A, a, b

1. HEIGHT
$$\boxed{A}=\frac{\text{height of } y}{\text{height of } x}$$

↓

2. MIRROR?
Mirrored $\Rightarrow \boxed{a}<0$.

↓

3. TWO EQUATIONS
$\boxed{a}\boxed{\tau_1}+\boxed{b}=\boxed{t_1}$ $\quad$ $\boxed{a}\boxed{\tau_2}+\boxed{b}=\boxed{t_2}$
(y at τ is x at $a\tau+b$)

↓

4. SOLVE
Subtract the equations for a. Back-substitute for b.

↓

5. ASSEMBLE, CHECK
$y(t)=\boxed{A}x(\boxed{a}t+\boxed{b})$. Test a third point.

7 — periodic signals

THE IDEA
A periodic signal is a song on repeat. Slide it by one period and nothing changes. The **fundamental period** is the shortest loop.

EXPLANATION

Pieces

Piece	Where to find it
ω	CT: the number multiplying t
$\hat{\omega}$	DT: the number multiplying n
m/N	$\hat{\omega}/2\pi$ as a fraction in lowest terms

Test	Means
$x(t+T)=x(t)$	CT periodic with period T
$x[n+N]=x[n]$, N an integer	DT periodic

Quick example	Work → answer
$\cos(\frac{4\pi}{6}n)$	$\frac{4}{12}=\frac{1}{3}\Rightarrow N=3$ reduce first
$\cos(n)$	$\frac{1}{2\pi}$ not rational: not periodic

Trap: A CT sinusoid is always periodic. A DT sinusoid is not always.
Trap: Reduce the fraction before reading N.

WORK

Steps — CT period

1. EACH TERM
$T_i = \dfrac{2\pi}{\boxed{\omega_i}}$

↓

2. RATIO (A SUM ONLY)
$\dfrac{T_1}{T_2} = \dfrac{p}{q}$ in lowest terms. Not a fraction of integers $\Rightarrow$ not periodic.

↓

3. COMBINE
$T = qT_1 = pT_2$ *(q loops of the first equal p loops of the second)*

Steps — DT period

1. DIVIDE BY 2π
$\dfrac{\boxed{\hat{\omega}}}{2\pi}$

↓

2. RATIONAL?
Not a fraction of integers $\Rightarrow$ not periodic. Stop.

↓

3. REDUCE
Write it as $\dfrac{m}{\boxed{N}}$ in lowest terms.

↓

4. READ N
The period is the denominator N. *(N photos hold m whole turns)*

↓

5. A SUM (DT)
$N = \text{lcm}(N_1, N_2, \ldots)$

8 — even and odd signals

THE IDEA
Even: a mirror on the vertical axis. **Odd**: spin the graph half a turn about the origin and it lands on itself. Every signal splits into one even part plus one odd part.

EXPLANATION

Pieces

Piece	Where to find it
even pile	terms with cos, t^2, $\lvert t\rvert$, constants
odd pile	terms with sin, t, t^3
$x(-t)$	replace every t with $-t$

Test	Result
$x(-t) = x(t)$	even
$x(-t) = -x(t)$	odd
neither holds	neither

Products work like signs: even·even = odd·odd = even. even·odd = odd.
Trap: An odd signal is 0 at the origin.
Trap: e^t and $u(t)$ fit neither pile. Use the formula steps.
Trap: DT: read $x[-n]$ from x at the opposite index.

WORK

Steps — test

1. FLIP
Write $x(-t)$. Simplify.

↓

2. COMPARE
Same as $x(t)$: even. Same with every sign flipped: odd. Else: neither.

Steps — split (sort the terms)

1. SORT THE TERMS
Each term goes in the even pile or the odd pile.

↓

2. READ THE PARTS
$\mathcal{E}v\{x\} =$ even pile $\qquad \mathcal{O}d\{x\} =$ odd pile

Steps — split (any formula)

1. FLIP
Write $x(-t)$.

↓

2. EVEN PART
$$\mathcal{E}v\{x\} = \frac{x(t) + x(-t)}{2}$$

↓

3. ODD PART
$$\mathcal{O}d\{x\} = \frac{x(t) - x(-t)}{2}$$

↓

4. CHECK
$\mathcal{E}v + \mathcal{O}d = x(t)$

Steps — split (any signal, DT table)

1. TABLE
Rows n, $x[n]$, $x[-n]$ for each n.

↓

2. EVEN PART
$$\mathcal{E}v = \frac{x[n] + x[-n]}{2}$$ in each column

↓

3. ODD PART
$$\mathcal{O}d = \frac{x[n] - x[-n]}{2}$$ in each column

↓

4. CHECK
$\mathcal{E}v + \mathcal{O}d = x[n]$ in each column. $\mathcal{O}d[0] = 0$.

9 — complex exponentials

THE IDEA
$e^{j\theta}$ is an arrow of length 1 at angle θ. Let $\theta = \omega t$ and it spins. Its shadow on the horizontal axis is a cosine. Multiply by $e^{\sigma t}$ and it also grows or shrinks: a spiral.

EXPLANATION

Pieces

Piece	Where to find it
$\lvert C\rvert$	length of C: $\sqrt{a^2+b^2}$
ϕ	angle of C: arctan $\frac{b}{a}$, check the quadrant
σ	real part of a: growth (+) or decay (−)
ω	imaginary part of a: spin rate

$x(t) = Ce^{at}, \quad C = \lvert C\rvert e^{j\phi}, \quad a = \sigma + j\omega.$

Case	Real part looks like
$\omega = 0$	real exponential: grows or decays
$\sigma = 0$	sinusoid, $T = \frac{2\pi}{\lvert\omega\rvert}$
both	damped sinusoid, envelope $\lvert C\rvert e^{\sigma t}$

DT: $x[n] = C\alpha^n$. $\lvert\alpha\rvert < 1$ decays, > 1 grows, $\alpha < 0$ flips sign each step. $e^{j(\hat{\omega}+2\pi)n} = e^{j\hat{\omega}n}$.

$C = a + jb$ **with**	**Angle** ϕ
$a > 0$	arctan $\frac{b}{a}$
$a < 0$	arctan $\frac{b}{a} + \pi$ *(left half: add π)*
$a = 0$	$\frac{\pi}{2}$ if $b > 0$, $\quad -\frac{\pi}{2}$ if $b < 0$

Trap: $e^{j2\pi k} = 1$ for every integer k. $\lvert e^{j\theta}\rvert = 1$.

WORK

Steps — Ce^{at} → real signal

1. POLAR C
$C = \boxed{\lvert C\rvert} e^{j\boxed{\phi}}$ $\qquad \lvert C\rvert = \sqrt{a^2+b^2}$, ϕ from the quadrant table

↓

2. SPLIT a
$a = \boxed{\sigma} + j\boxed{\omega}$

↓

3. REAL PART
$\mathrm{Re}\{Ce^{at}\} = \boxed{\lvert C\rvert} e^{\boxed{\sigma}t} \cos\left(\boxed{\omega}t + \boxed{\phi}\right)$

Steps — real signal → Ce^{at}

1. MATCH
Read $\lvert C\rvert$, σ, ω, ϕ off $\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi)$.

↓

2. ASSEMBLE
$C = \boxed{\lvert C\rvert} e^{j\boxed{\phi}}$ $\qquad a = \boxed{\sigma} + j\boxed{\omega}$

Steps — cosine or sine → two exponentials

1. EULER
$\cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2}$ $\qquad \sin\theta = \dfrac{e^{j\theta} - e^{-j\theta}}{2j}$

↓

2. PLACE
$\boxed{A}\cos(\boxed{\omega}t) = \dfrac{\boxed{A}}{2} e^{j\boxed{\omega}t} + \dfrac{\boxed{A}}{2} e^{-j\boxed{\omega}t}$

10 — the button, the switch, and the ramp

THE IDEA
δ is a **button**: pressed once at time 0. u is a **switch**: flipped on at 0 and left on. r is a **ramp**: the running total of the switch. Each is the running total of the one before. Everything else is built from these three.

Where is it? Stems as buttons, pulses as switches

EXPLANATION

Pieces

Piece	Where to find it
h	the height of a stem (the coefficient)
k	the position: set the inside to 0
a, b	switch on at a, off at b

Inside	Position, direction
$n-3$	at $n=3$: delay, right
$n+2$	at $n=-2$: advance, left

Given → find	Do this
$u \to \delta$ (CT) / (DT)	$\frac{du}{dt}$ / $u[n]-u[n-1]$
$\delta \to u$ (CT) / (DT)	$\int_{-\infty}^{t} \delta$ / $\sum_{k\le n} \delta[k]$

Trap: The off-switch's own position is already off: $u[n]-u[n-4]$ is 0 at $n=4$.
Trap: $u[n]$ is $0,0,1,1,1,1,\ldots$ It does not climb. The ramp climbs.

WORK

Steps — stems → buttons

1. READ EACH STEM
height h at position k

↓

2. ONE BUTTON PER STEM
$x[n] = \sum \boxed{h}\,\delta[n-\boxed{k}]$ *(no stem, no term)*

Steps — buttons → stems

1. EACH TERM
coefficient = height h inside $=0 \Rightarrow$ position k

Steps — pulse from switches

1. TEMPLATE
DT: $u[n-\boxed{a}]-u[n-\boxed{b}]$ is on for $\boxed{a} \le n \le \boxed{b}-1$
CT: $u(t-\boxed{t_1})-u(t-\boxed{t_2})$ is on for $\boxed{t_1} \le t < \boxed{t_2}$

↓

2. HEIGHT
Multiply the pair by the pulse height.

Sifting: the camera flash

EXPLANATION

Pieces

Piece	Where to find it
t_0	the flash: set the inside of δ to 0
lo, hi	the limits of the integral (the window)
$g(t)$	everything multiplying δ

CT button: area 1 (not height). $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$.
No integral (sampling form): $g(t)\,\delta(t-t_0) = g(t_0)\,\delta(t-t_0)$. The δ stays.

Quick example	Work → answer
$\int_0^5 e^t\,\delta(t+1)\,dt$	flash -1, outside $[0,5]$: 0
$\sum_n n^2\,\delta[n-3]$	flash 3, all n: 3^2

Trap: The answer has no δ left in it.

WORK

1. WHERE IS THE FLASH?
Set the inside of δ to 0: $t = \boxed{t_0}$

↓

2. IN THE WINDOW?
Is $\boxed{t_0}$ in $[\boxed{\text{lo}}, \boxed{\text{hi}}]$? No $\Rightarrow$ the answer is 0.

↓

3. WHAT DOES IT CAPTURE?
$\int \boxed{g(t)}\,\delta(t-\boxed{t_0})\,dt = \boxed{g}(\boxed{t_0})$ *(plug t_0 into everything except δ)*

Graphs from ramps and switches

EXPLANATION

Pieces

Piece	Where to find it
corner	where the slope changes
Δslope	slope after $-$ slope before
jump	where the graph jumps, and by how much (up $+$)

Form	Formula
ramp	$r(t) = t\,u(t), \quad r[n] = n\,u[n]$
CT chain	$\frac{dr}{dt} = u, \quad \frac{du}{dt} = \delta$
DT chain	$r[n] - r[n-1] = u[n-1]$
DT running sum	$r[n] = \sum_{k=-\infty}^{n-1} u[k]$

Trap: The DT ramp sum stops at $n-1$.
Trap: At a peak the slope change is double: $+1 \to -1$ is -2.

WORK

Steps — graph $\to$ formula

1. WALK LEFT TO RIGHT
List every corner and every jump.

↓

2. SLOPE CHANGE AT EACH CORNER
$\boxed{\Delta}$ = slope after $-$ slope before

↓

3. ASSEMBLE
$x(t) = \sum \boxed{\Delta} r(t - \boxed{\text{corner}}) + \sum \boxed{\text{jump}}\, u(t - \boxed{\text{where}})$

↓

4. CHECK
The slope changes sum to 0 if the graph ends flat at 0.

Steps — formula $\to$ graph

1. EACH RAMP IS A CORNER
$\boxed{\Delta} r(t - \boxed{\text{corner}})$: the slope changes by Δ there.

↓

2. EACH SWITCH IS A JUMP
$\boxed{\text{jump}}\, u(t - \boxed{\text{where}})$

↓

3. DRAW
Start flat at 0. Add each change in time order.

Steps — derivative of a graph

1. EACH PIECE
A piece with slope m becomes a flat level m. *(ramps become switches)*

↓

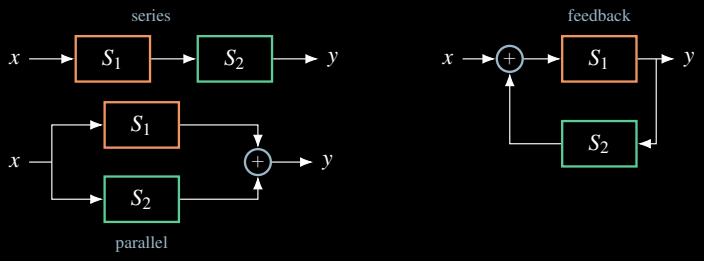
2. EACH JUMP
A jump becomes a button: $\boxed{\text{jump}}\, \delta(t - \boxed{\text{where}})$. *(switches become buttons)*

11 — systems and interconnections

THE IDEA
A **system** is a machine: a signal goes in, another comes out. A **difference equation** is its recipe: how to cook $y[n]$ from inputs, and sometimes from its own past outputs.

Hookups: series, parallel, feedback

EXPLANATION



Pieces

Piece	Where to find it
S_1	the first machine (or top branch)
S_2	the second machine (or bottom branch, or the loop)
start	a recursive system's start value, $y[-1] = 0$ at rest

Trap: Shifting $w[n] = 2x[n]$ to $w[n-1]$ shifts every n: $2x[n-1]$.
Trap: Two LTI systems in series can swap order. Other systems cannot.

WORK

Steps — series

1. NAME THE MIDDLE
$w =$ output of S_1.

↓

2. SHIFT AS NEEDED
Write $w[n-1]$, $w[n-2]$, … by replacing every n in S_1.

↓

3. SUBSTITUTE
Put w into S_2's input slot. The answer has only x and y.

Steps — parallel

1. ADD THE BRANCHES
$y = \boxed{y_1} + \boxed{y_2}$ *(both get the same x)*

Steps — run a nonrecursive system

1. LIST THE INPUT
Write each nonzero $x[n]$. Every other x is 0.

↓

2. EACH OUTPUT TIME
For each n, look up every $x[\cdot]$ the recipe needs, and combine. *(y is nonzero from (first input + smallest delay) to (last input + largest delay))*

Steps — run a recursive system

1. START AT REST
$y[-1] = \boxed{0}$ (or the given start).

↓

2. ONE STEP AT A TIME
$y[0]$ from $x[0]$ and $y[-1]$. Then $y[1]$ from $x[1]$ and $y[0]$. …

↓

3. PATTERN
Spot the rule (often $c\,a^n u[n]$). Check one more n.

Diagram ↔ equation

EXPLANATION

Pieces

Block	Does
adder (circle)	sums its inputs
gain (box a)	a·input
unit delay (box D)	$x[n] \to x[n-1]$
branch dot	copies a signal

System	Equation
first difference	$y = x[n] - x[n-1]$
3-pt moving average	$y = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$
accumulator	$y[n] = y[n-1] + x[n]$

Nonrecursive (FIR): inputs only. **Recursive (IIR)**: also past outputs. Needs a start value.
Trap: A D that feeds y back into the adder means $y[n-1]$ in the equation.

WORK

Steps — diagram → equation

1. NAME EACH WIRE
After k delays, a wire carries $x[n-k]$ or $y[n-k]$.

↓

2. EACH ADDER
output = sum of the wires in, each times its gain.

↓

3. CHECK THE COUNT
One D per shift. One term per wire into the adder.

Steps — equation → diagram

1. ONE BLOCK PER PIECE
Each $x[n-k]$: k delays on the input side. Each $y[n-k]$: k delays on the output side (feedback).

↓

2. GAINS
One gain box per coefficient.

↓

3. ADDER
All terms meet in one adder. Its output is $y[n]$.

↓

4. READ IT BACK
Diagram → equation must give the original.

12 — system properties

THE IDEA
Properties are yes/no questions about a machine. Each is separate: a machine can pass one and fail another.

Memory, causality, invertibility

EXPLANATION

Pieces: the time stamps

Stamp on $x[\cdot]$	Color
n (or t)	now
$n-1, n-2, \ldots$	past
$n+1, \ldots,$ or $-n, 2n$	future (for some n)

CT works the same with t: $x(t-1)$ is past. $x(t+1)$, $x(2t)$, $x(-t)$ reach the future for some t.
Trap: $y(t) = x(-t)$: at $t = -1$ it needs $x(1)$, the future.
Trap: $y = x^2$ is not invertible: 1 and -1 both give 1.

Common inverses

System	Inverse
$y = ax[n], \quad a \neq 0$	$x[n] = \frac{1}{a}y[n]$
$y = x[n - n_0]$	$x[n] = y[n + n_0]$
accumulator	first difference: $x[n] = y[n] - y[n-1]$
$x^2, \quad \lvert x\rvert, \quad x[2n], \quad 0 \cdot x$	none: information is lost

WORK

1. COLOR EACH STAMP
now, past, or future.

↓

2. MEMORYLESS?
Every stamp is $\boxed{\text{now}}$ ⇒ memoryless. Else it has memory.

↓

3. CAUSAL?
No stamp is $\boxed{\text{future}}$ ⇒ causal.

↓

4. INVERTIBLE?
Solve for x. Exactly one answer ⇒ invertible. *(find two inputs with the same output to show no)*

Linearity: mix before the machine, or after?

EXPLANATION
Linear means: mixing two inputs *before* the machine gives the same output as running each one alone and mixing the outputs *after*. Mixing means $c_1x_1 + c_2x_2$ for any numbers c_1, c_2.

Pieces

Piece	Where to find it
$\square$	the input box: every place the recipe has an x. Keep the index on it: $\square[n-1]$
x_1, x_2	two test inputs. Never pick numbers: keep them as letters
c_1, c_2	the knobs: any two numbers
y_1, y_2	the recipe run on x_1 alone, and on x_2 alone

Shortcut: look at how x enters each term

Term in the recipe	Linear?
a number or a function of n times one x: $3x[n], \quad nx[n], \quad \frac{x[n]}{n}, \quad e^n x[n]$	yes
x shifted, flipped, or squeezed: $x[n-1], \quad x[-n], \quad x[2n]$	yes
x inside a function: $x^2, \quad \sqrt{x}, \quad e^x, \quad \lvert x\rvert, \quad \cos(x[n])$	no
two x's multiplied: $x[n]\,x[n-1]$	no
a term with no x: $+3, \quad +\sin n, \quad +u[n]$	no (fails the zero test)

Every term yes ⇒ linear. One no ⇒ not linear. The steps on the right prove it.

Quick example	Work → answer
$y = nx[n]$	P1: $n(c_1x_1 + c_2x_2)$ P2: $c_1nx_1 + c_2nx_2$ equal
$y = x[n]^2$	P1: $c_1^2x_1^2 + 2c_1c_2x_1x_2 + c_2^2x_2^2$ P2: $c_1x_1^2 + c_2x_2^2$ not equal
$y = 2x[n] + 3$	zero test: $2(0) + 3 = 3 \neq 0$ not linear

Trap: The coefficients may depend on n. Only how x appears matters.
Trap: Use knobs c_1, c_2, not a, b. The system may have its own a.
Trap: Passing the zero test alone does not prove linear.

WORK

1. BOX THE INPUT
Rewrite the recipe with an empty box wherever it says x. Keep each index.
$y[n]$ = recipe of $\square$ for example $n \cdot \square[n] + \square[n-1]$

↓

2. ZERO TEST
Put 0 in every box. If $y \neq 0$ ⇒ not linear. Stop.

↓

3. PATH 1: MIX, THEN MACHINE
Put $(\boxed{c_1}\boxed{x_1} + \boxed{c_2}\boxed{x_2})$ in every box, with that box's index.
Expand: multiply out every bracket. Call it **P1**.

↓

4. PATH 2, PART A: MACHINE ON EACH
$\boxed{y_1}$ = recipe with x_1 in every box.
$\boxed{y_2}$ = recipe with x_2 in every box.

↓

5. PATH 2, PART B: THEN MIX
P2 = $\boxed{c_1}\boxed{y_1} + \boxed{c_2}\boxed{y_2}$. Expand.

↓

6. COMPARE TERM BY TERM
Match each term of P1 with a term of P2.
All match, for every c_1, c_2 ⇒ linear.
A leftover term ($c_1^2, \quad c_1c_2x_1x_2, \quad$ an extra constant) ⇒ not linear. The leftover is your reason.

Time invariance: delay before the machine, or after?

EXPLANATION

Time invariant means: delaying the input by n_0 just delays the output by n_0. The machine ignores the clock.

Pieces

Piece	Where to find it
$\square$	the input box: every place the recipe has an x
inside of x	the index in each box: n, $n-1$, $2n$, ...
lone n	any n (or t) outside a box: $nx[n]$, $\frac{x[n]}{n}$
n_0	the delay you apply

Shortcut: look for the clock

Recipe has	Time invariant?
only shifts inside: $x[n-1]$, $x[n+2]$	yes
a lone n: $nx[n]$, $\cos(n)\,x[n]$	no
a squeeze or flip inside: $x[2n]$, $x[-n]$	no
numbers only outside: $3x[n]$, x^2	yes

Trap: Path 1 changes only the inside of x. Path 2 changes every n.

WORK

1. BOX THE INPUT
Rewrite the recipe with $\square[\boxed{\text{inside}}]$ wherever it says x. Mark each lone n.

↓

2. PATH 1: DELAY, THEN MACHINE
In each box only: $\boxed{\text{inside}} \to \boxed{\text{inside}} - \boxed{n_0}$.
Leave every lone n alone.

↓

3. PATH 2: MACHINE, THEN DELAY
Replace every n with $n - \boxed{n_0}$: inside the boxes *and* each $\boxed{\text{lone } n}$.

↓

4. COMPARE
Same ⇒ time invariant. Different ⇒ time varying.

Stability with a ceiling B

EXPLANATION

Pieces

Piece	Where to find it
$\lvert c_i\rvert$	the size of each coefficient
B	the input ceiling: $\lvert x[n]\rvert \le B$

BIBO: bounded in ⇒ bounded out. **LTI shortcut**: stable $\iff \sum\lvert h[n]\rvert < \infty$ (topic 13).

Red flag	Why
$nx[n]$	$\lvert n\rvert B$ grows with n
accumulator	adds another B every step
$y = a\,y[n-1] + x$, $\lvert a\rvert \ge 1$	the echo never shrinks

WORK

1. ABSOLUTE VALUE
Take $\lvert\cdot\rvert$ of both sides.

↓

2. SPLIT
$\lvert a+b\rvert \le \lvert a\rvert + \lvert b\rvert$, $\lvert cx\rvert = \lvert c\rvert\lvert x\rvert$

↓

3. CAP EACH INPUT
$\lvert y\rvert \le \boxed{\lvert c_1\rvert}\boxed{B} + \boxed{\lvert c_2\rvert}\boxed{B} + \cdots$ *(shifted x too)*

↓

4. VERDICT
A finite number (no n, no endless sum) ⇒ stable.
To show unstable: one capped input ($x = 1$, $u[n]$) with an unbounded output.

13 — the convolution sum

THE IDEA

Every DT input is a stack of buttons. Press one button and an LTI machine answers with its echo, the **impulse response** $h[n]$. Each input stem makes its own echo, scaled by its height and shifted to its time. The output is the sum of all the echoes: $y = x * h$.

Find h, and read the system from h

EXPLANATION

Pieces

Piece	Where to find it
b_k	the coefficient of $x[n-k]$
k	the delay of that term
$x = \delta$	the test input: 1 at $n = 0$, else 0

From h	Property
$h[n] = 0$ for $n < 0$	causal
$\sum \lvert h[n] \rvert < \infty$	BIBO stable
$h = c\,\delta[n]$	memoryless
finitely many stems	FIR (nonrecursive)
series / parallel	$h_1 * h_2$ / $h_1 + h_2$

Trap: Only for LTI systems at rest. $y = x^2$ has no h.

WORK

Steps — nonrecursive → h

1. READ THE COEFFICIENTS

$y = \sum \boxed{b_k}\, x[n - \boxed{k}]$

↓

2. ONE STEM EACH

$h[n] = \sum \boxed{b_k}\, \delta[n - \boxed{k}]$

Steps — recursive → h

1. PRESS THE BUTTON

Put $\boxed{x = \delta}$. Start at rest: $h[-1] = 0$.

↓

2. RUN IT

$h[0]$, $h[1]$, $h[2]$, … one step at a time.

↓

3. PATTERN

Usually $h = a^n u[n]$.

Steps — h → equation

1. ONE TERM PER STEM

$y[n] = \sum_k h[k]\, x[n-k]$

Convolve x and h

EXPLANATION

Pieces

Piece	Where to find it
$x[k]$	each input stem (Method 1: one color per stem)
h	the echo, as stems or a formula
n	the output time. Fixed during one sum

Fact	Formula
convolution sum	$y[n] = \sum_k x[k]\,h[n-k]$
swap	$x * h = h * x$
x on $[a,b]$, h on $[c,d]$	y on $[a+c,\ b+d]$
length	$L_x + L_h - 1$
sum check	$\sum y = (\sum x)(\sum h)$
$x * \delta[n-n_0]$	$x[n-n_0]$
$a^n u * b^n u$, $a \neq b$	$\dfrac{a^{n+1} - b^{n+1}}{a-b} u[n]$
$a^n u * a^n u$	$(n+1)a^n u[n]$
finite geometric	$\sum_{k=0}^{n} r^k = \dfrac{1-r^{n+1}}{1-r}$

If the signals are	Use
two short lists of stems	Method 1, table
short pulse, long formula	Method 2, flip and slide
two formulas with switches	Method 3, formula

Trap: $h[n-k]$, not $h[k-n]$. And $y \neq x \cdot h$.
Trap: After the flip, $h[0]$ sits at $k = n$, and $h[1]$ at $k = n-1$.

WORK

Method 1 — the table

1. RANGE AND LENGTH
y on $[a+c,\ b+d]$, length $L_x + L_h - 1$.

2. ONE ROW PER INPUT STEM
row k: $\boxed{x[k]} \cdot \boxed{h}$, starting at column $n = k$

3. ADD EACH COLUMN
$y[n]$ = sum of column n

4. CHECK
$\sum y = (\sum x)(\sum h)$

Method 2 — flip and slide

1. DRAW $x[k]$
on a k axis

2. FLIP AND SLIDE h
$h[k] \to h[-k] \to h[\boxed{n} - k]$. Its first stem sits at $k = n$, the rest trail left.

3. OVERLAP LIMITS
x on $[a,b]$, h on $[c,d]$: the sum runs
$k = max(a,\ n-d)$ to $min(b,\ n-c)$ *(h never ends: $d = \infty$, so the lower limit is a)*
Lower limit $>$ upper limit $\Rightarrow$ no overlap, $y = 0$.

4. REGIONS
A new region of n starts each time a different bound wins the max or the min: none, partial, full, (partial, none).

5. EACH REGION
$$y[\boxed{n}] = \sum_{k \in \text{overlap}} \boxed{x[k]}\,\boxed{h[n-k]}$$

6. CHECK
The region formulas agree where they meet.

Method 3 — formula

1. PLACE
$$y[n] = \sum_k \boxed{x[k]}\,\boxed{h[n-k]}$$

2. SWITCHES CUT THE SUM
$u[k]\,u[n-k] \Rightarrow k = 0..\boxed{n}$, and $y = 0$ for $n < 0$.

3. PULL OUT, GEOMETRIC SUM
$a^k\,b^{n-k} = b^n \left(\dfrac{a}{b}\right)^k$: pull out b^n, ratio $r = \dfrac{a}{b}$.
$$\sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r}$$ *($r = 1$: the sum is $n+1$)*

4. CHECK
$y[0] = x[0]\,h[0]$.